Promoting Natural Gait Cycle to Foot Drop Patients

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Why drop foot?

- Immediately stood out as interesting topic of interest.
- Team was already considering ideas during project discussion
- Current recovery methods include invasive surgery

Other project considerations included…

- Back/Spinal Prosthesis
- Tinnitus Muter
- Hypermobility Brace
- Medical Simulation Device for non-humans

Background on Drop Foot

- Drop foot can be seen in patients with neurologic, spinal, autoimmune, and musculoskeletal disorders [1].
- The peroneal nerve is damaged and loss of function ensues.
- Gait cycle is affected and increased fall incidence.

- About 19 out 100,000 have Fibular Neuropathy [2].
- About 2 out of 100,000 have ALS [2].
- About 1 out of 100,000 have AIDP (Polyneuropathy) [2].

Background on Drop Foot (cont.)

- Many people recover without surgical intervention, some cases are severe and lead to permanent loss of function.
- Surgical intervention has about a 70% success rate.
- Long-term effects of drop-foot: Weakness and decrease in muscle mass, worsening balance, and issues with mobility
- Recovery can look like a brace, physical therapy, nerve stimulation, and surgery in extreme cases

Literature Review

The Condition

- Drop-foot is the inability to lift the forefront of the foot and leads to the toes being dragged when walking
    - Weakness or paralysis in foot muscles
- Steppage gait affect: A person may compensate for toe-dragging by lifting their knee higher
    - They might swing their leg instead
    - Both increase the odds of tripping and falling

The Causes

- Injuries:
    - Ankle fracture
    - Fibula fracture
    - Knee dislocation
    - Knee fracture

- Lumbar radiculopathy: When a nerve is pinched in the lower back [7]
    - Primarily the sciatic nerve that runs down the leg to the foot
    - Five vertebrae in your back, typically L5 LR occurs the most, followed by L4 and sacral regions

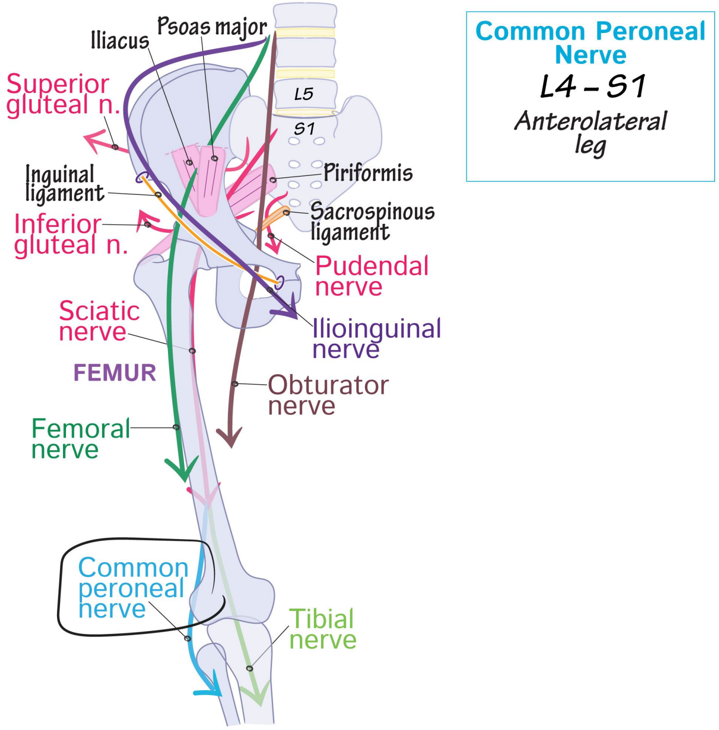


Figure 1. Location of nerves in the lower back, hip, and leg [8]

Literature Review

Diagnosis

- Drop-foot can be diagnosed through a physical exam or go more in depth with electromyography (EMG) as well as X-rays or an MRI
  - EMG can be getting a needle inserted into the muscle of the affected area and testing movement to see if proper electrical activity is present [6]

Treatment

- Treatment can range from physical therapy and orthotic devices to surgery, depending on severity

- Early intervention and recovery can reduce time to heal and prevent irrecoverable damage
  - Muscle twitching can be a positive sign, indicating neural recovery

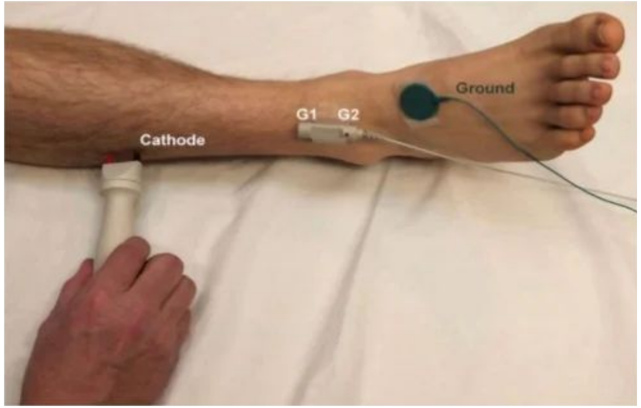


Figure 2. EMG being performed on a person to test for drop-foot [9]

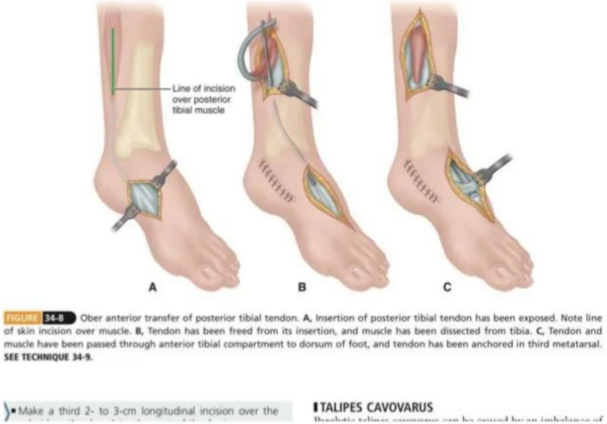


Figure 3. Illustration of surgery being carried out on a person with drop-foot [10]

Current Device Issues

- Braces are stiff
- Limit range of motion, either dorsiflexion or plantarflexion
- Most are non-discrete
- Can cause gait cycle / movement modifications
    - Can cause long term injury to patient
- Powered devices exist only in research

Figure 4. Current market brace standard [11]

Stakeholders

- Primary
  - Drop-foot patients
  - Orthotists
  - Medical Organizations
  - Clinic Practices
- Secondary
  - Manufacturers
  - Physical therapists
  - Families
  - Regulatory Figures
  - Policymakers
  - Insurance Companies

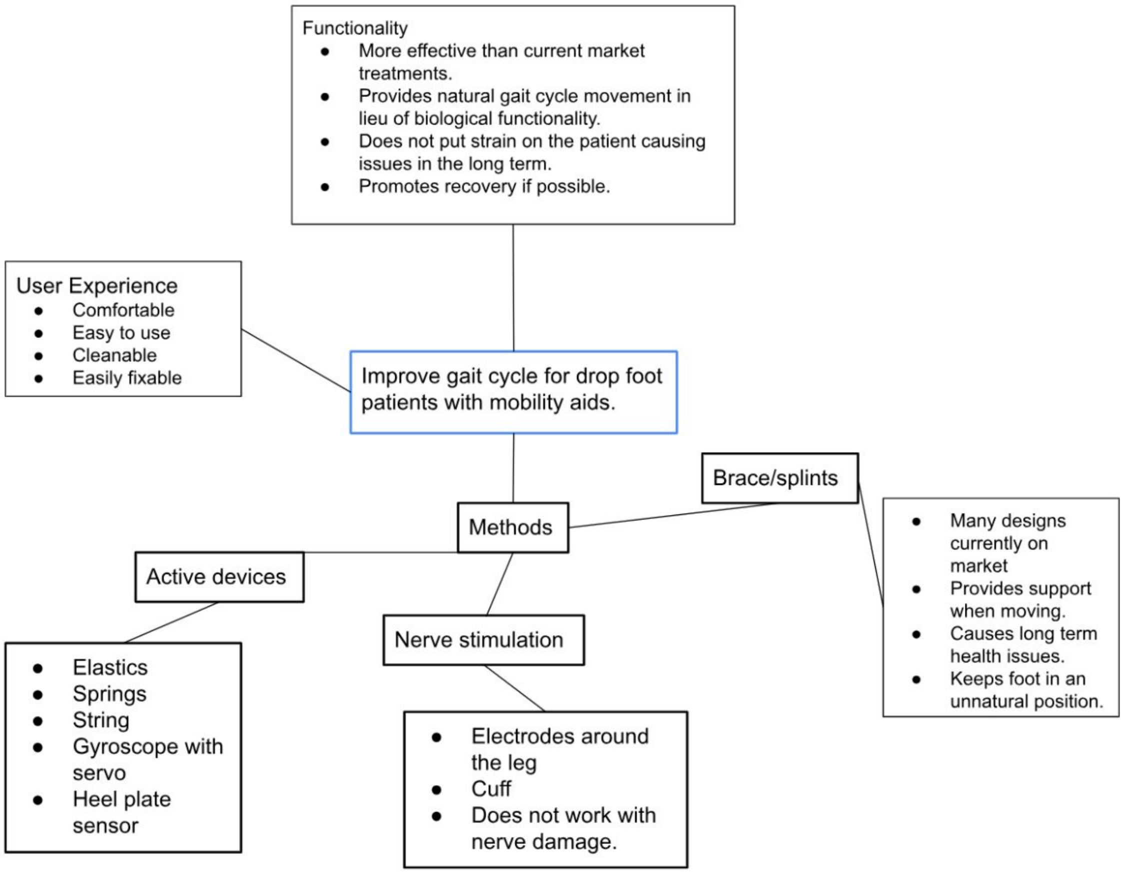
List of questions for O&P Clinics

1. How familiar are you with drop foot?
2. How does drop foot affect the lives of your patients?
3. What issues do you believe exist with existing devices?
4. What issues do your patients express with existing devices?
5. What feature(if any) do you believe would improve patient recovery or quality of life?
6. Do your patients express a preference to a device that could be worn within a shoe or without?

Project Statement

Function cost-effectiveness. The goal of the drop foot device is to improve its functionality through a more natural gait cycle. It should be kept within a price-effective range while providing noticeable improvements.

Requirements (Concept map)



Design requirements

1. Performance
- Sustain a travel speed of 2 mph
- Dorsiflex between 15-30° and plantarflex between 20-50°
- Provide just above the minimal amount of force to lift the front of the foot

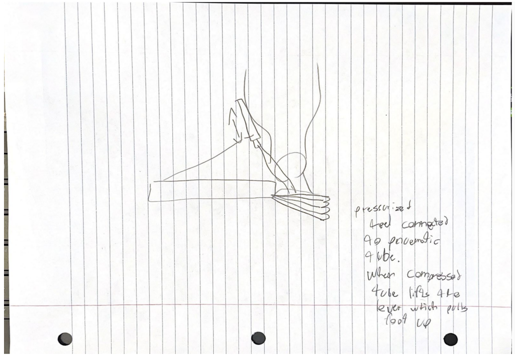
2. Comfort and Usability
- Comfortable to encouraging use
- Light as possible (2 lbs) to minimize energy expenditure
- Easy to put on and take off
- Wearable for long periods (8-10 hours)
- Non-irritable

Design requirements

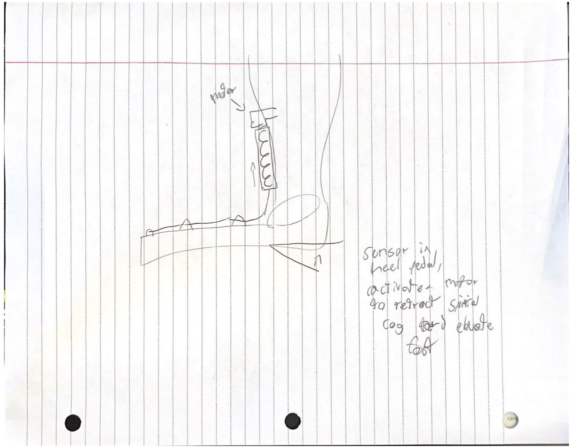
3. Noise level
   - Operate quietly (30-40 dB) to minimize disturbance
   - Not exceed 60 dB to disturb walking conversation

4. Durability and maintenance
   - Weather-resistant
   - Manufacturable and fit for mass production
   - Require little maintenance
   - Maintain a natural gait cycle
   - Cleanable
   - Adjustable

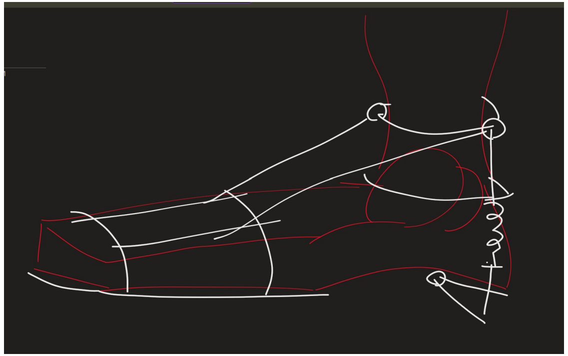
Preliminary Design Concepts



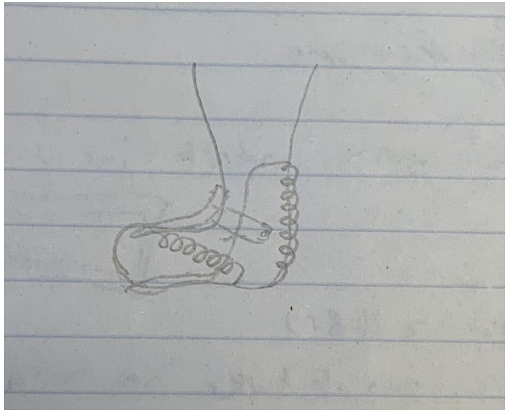
Pneumatic Pump - CHS



Corkscrew Motor with Heel Plate - CHS

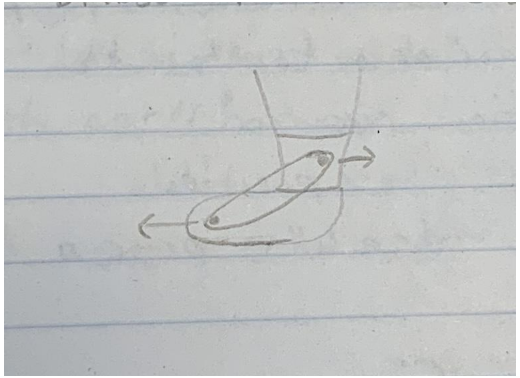


Spring Based Tensioners - CHS

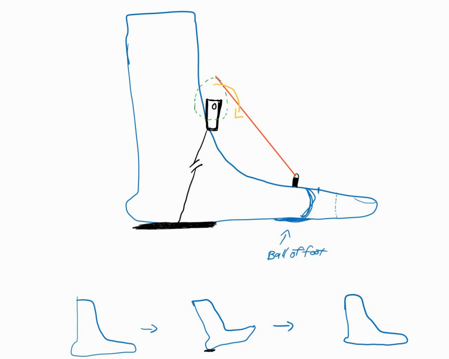


Two Spring Design - GMS

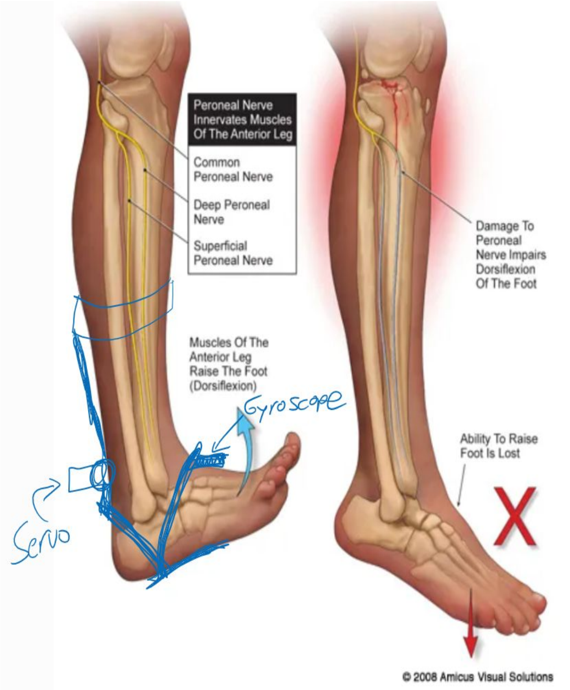
Preliminary Designs Cont.



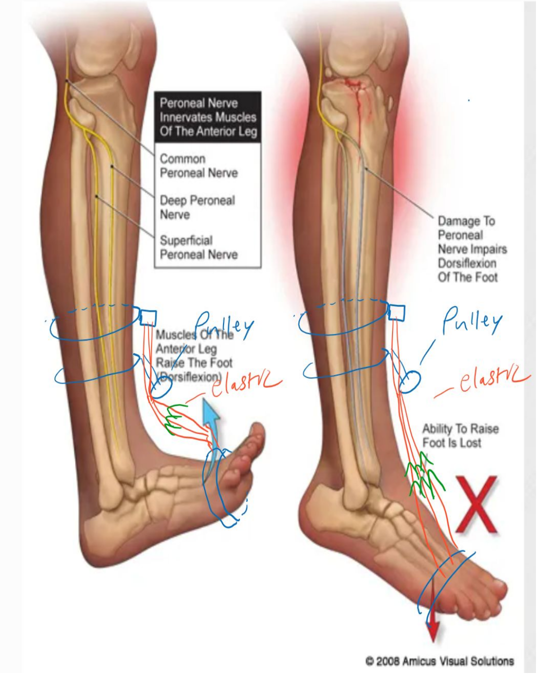
Elastic with Moving Tensioner - GMS



Heel Strike Servo Rotator - EI



Gyroscopic Heel Actuator - EI



Elastic Based Pulley System - EI

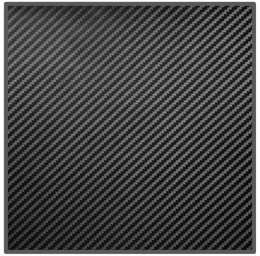
https://www.mypanotes.co.uk/peroneal-nerve-injury-foot-drop

Material Considerations

- Lightweight
- Durable
- Flexible
- Cost-effective
- Rust-proof
- Weather-resistant
- Temperature-resistant
- Anti-bacterial

Polylactic Acid (PLA)

Aluminum



Carbon Fiber

Why elastics?

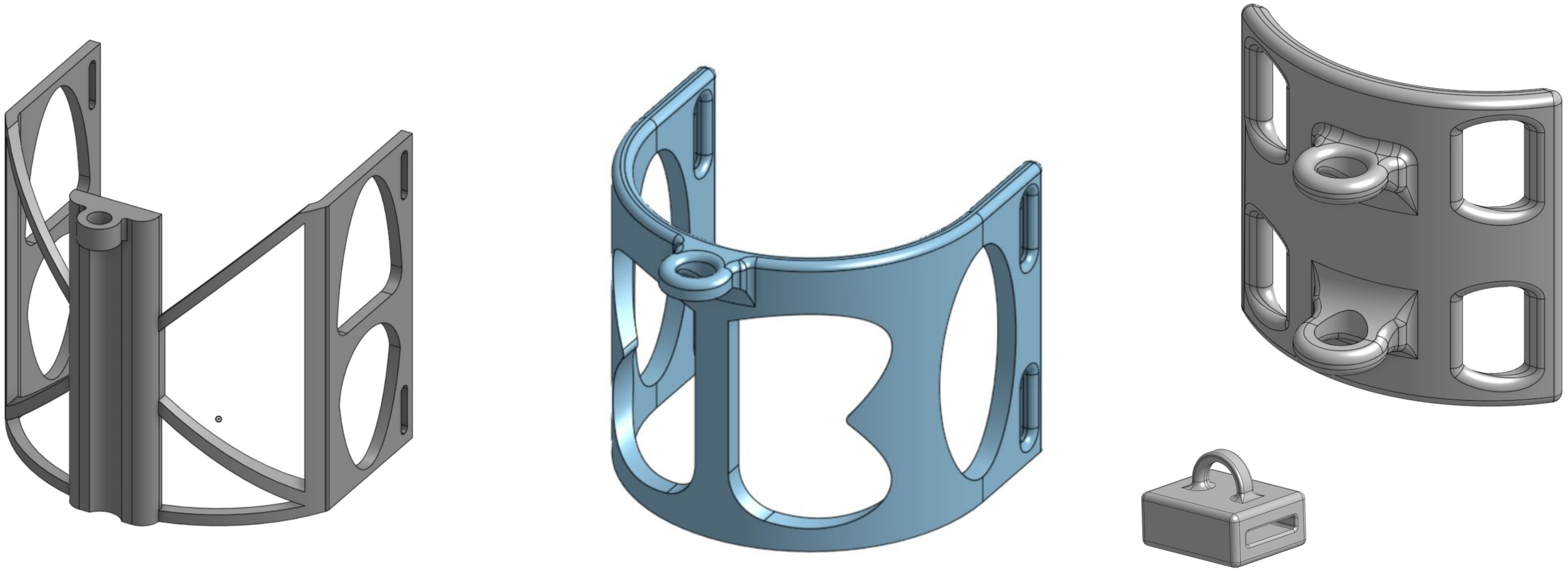
- Elastics promote a more dynamic walking pattern allowing dorsiflexion and plantarflexion of the foot
- Mechanically works most similar to a person’s tendon
- For repairable damaged nerves, elastics will strengthen current neural connections and create new connections in the foot - neuroplasticity

The Stop-Drop


BROOKS

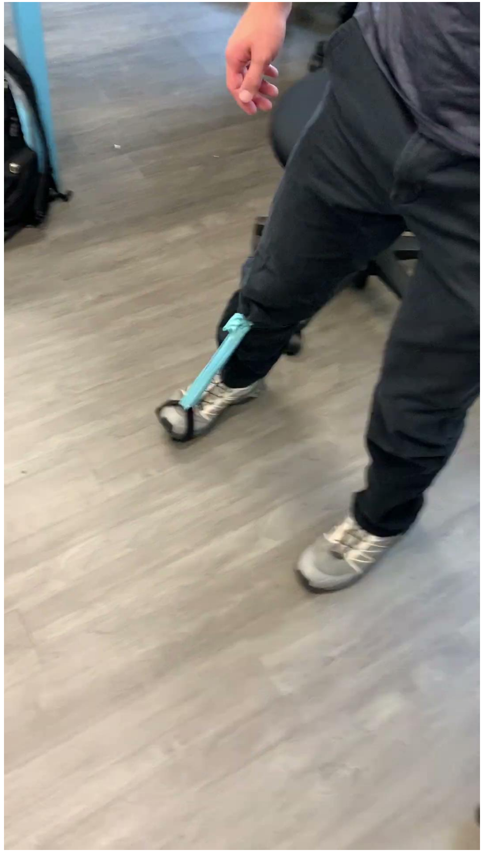
Drop foot subject baseline

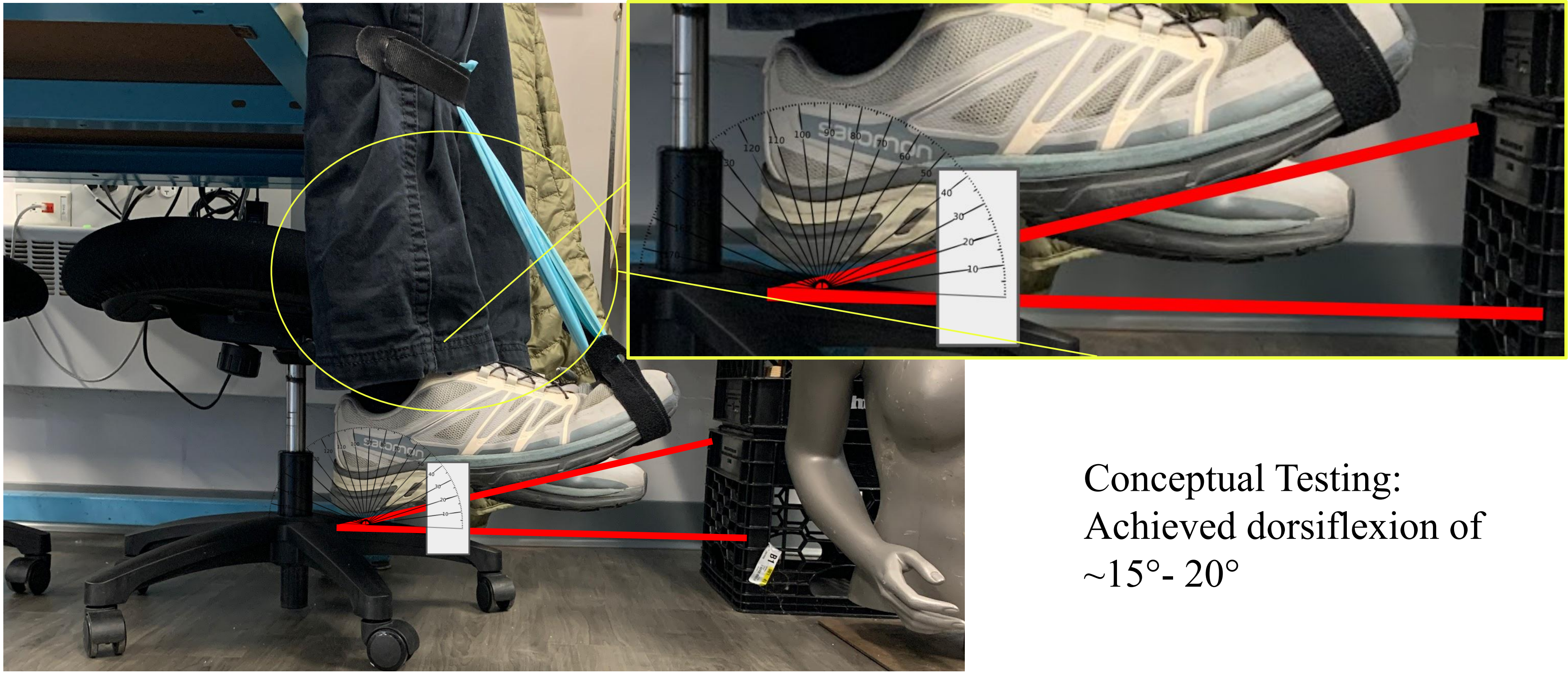
Iterations of Works-Like Models



V1 V2 V3

Works-Like Model

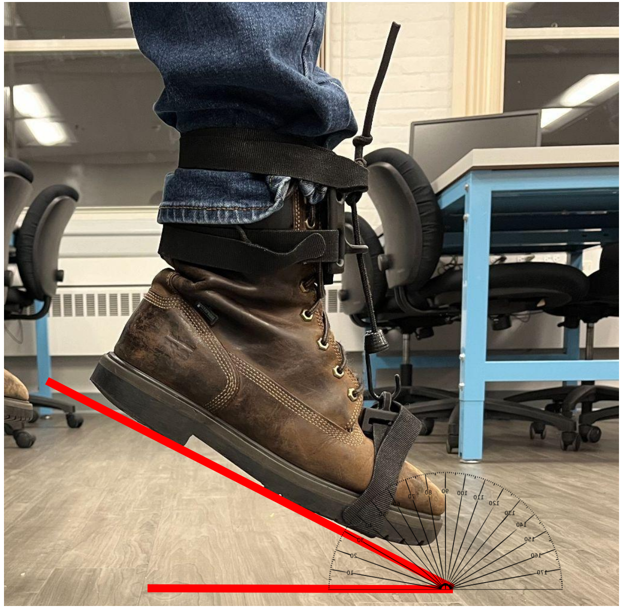




Conceptual Testing:
Achieved dorsiflexion of
~15°- 20°

V3 Flexion Achievements

~ 5° of dorsiflexion



~ 30° of plantarflexion
(with applied downward force
for demonstration)

V2 On foot testing

V3 On foot testing

Testing V3

Is-Like Model (Future Development)

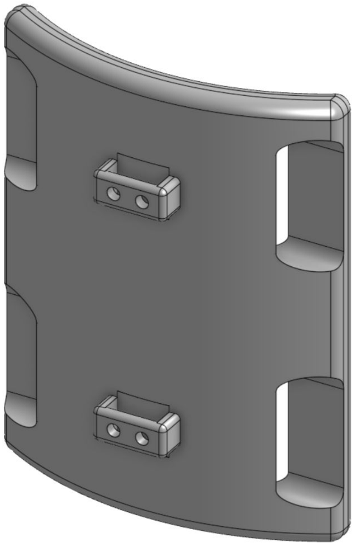
V4 Concepts

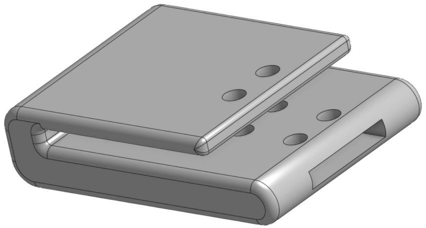
The final product may be made of a flexible metal like aluminum, or of a composite like carbon fiber.

Further testing could prove that a flexible structure is viable for comfort and usability.

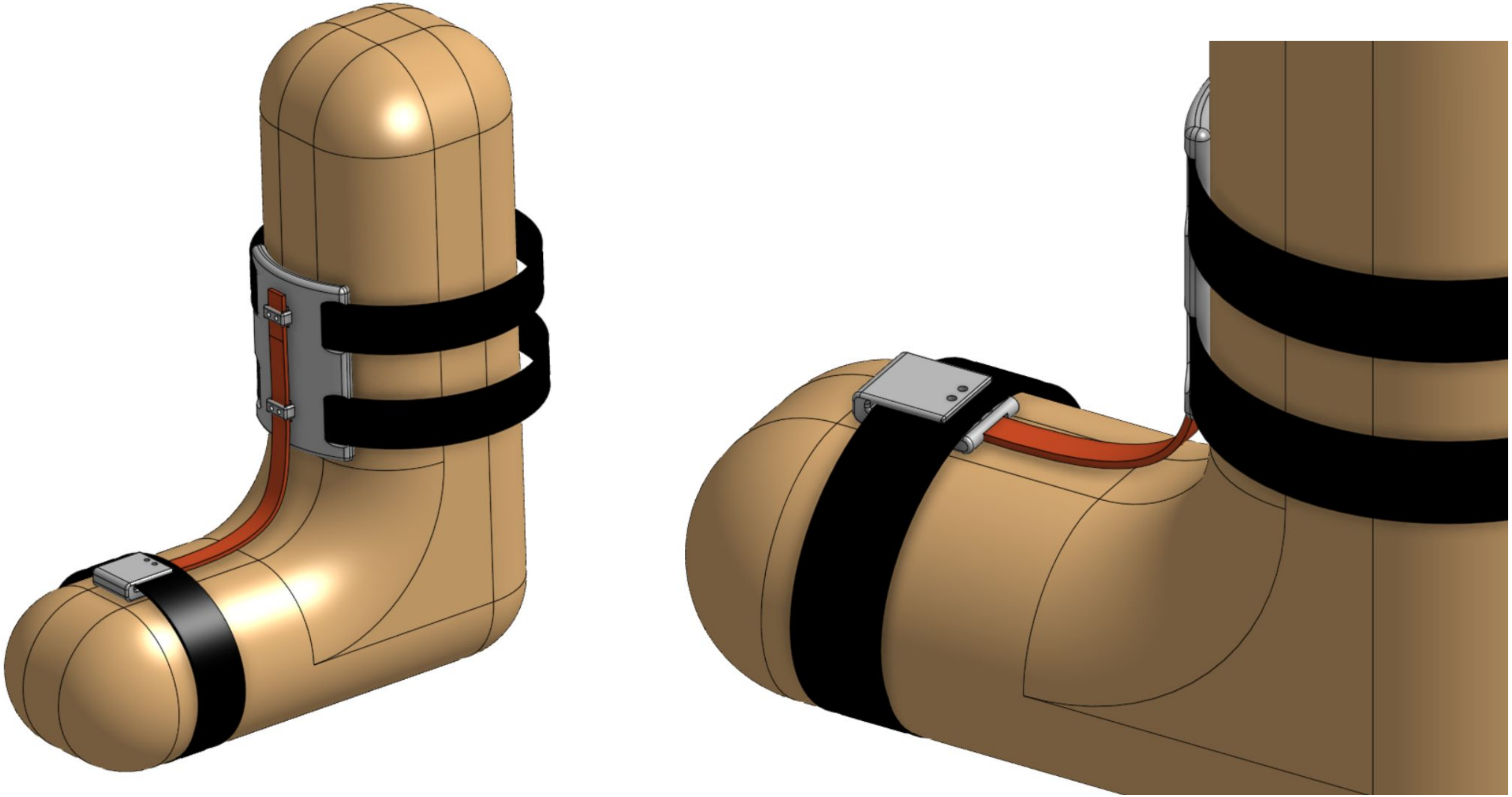
Slim profile "Sock Clip" will fit inside the

shoe and will attach to a flat elastic band.



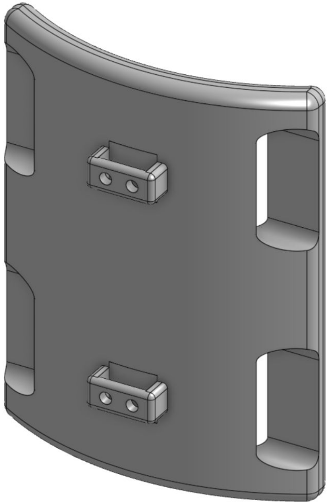


Not to relative scale



[12]

Features of the Final Design



Elastics

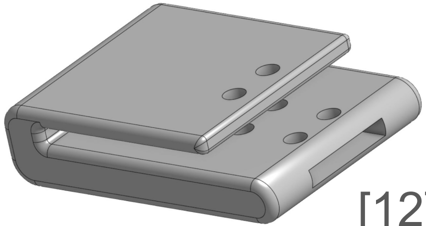
- Durable
- Appropriate tension
- Low Profile

Material Selection

- Durable
- Lightweight
- Flexible

Velcro

- User friendly
- Cost effective

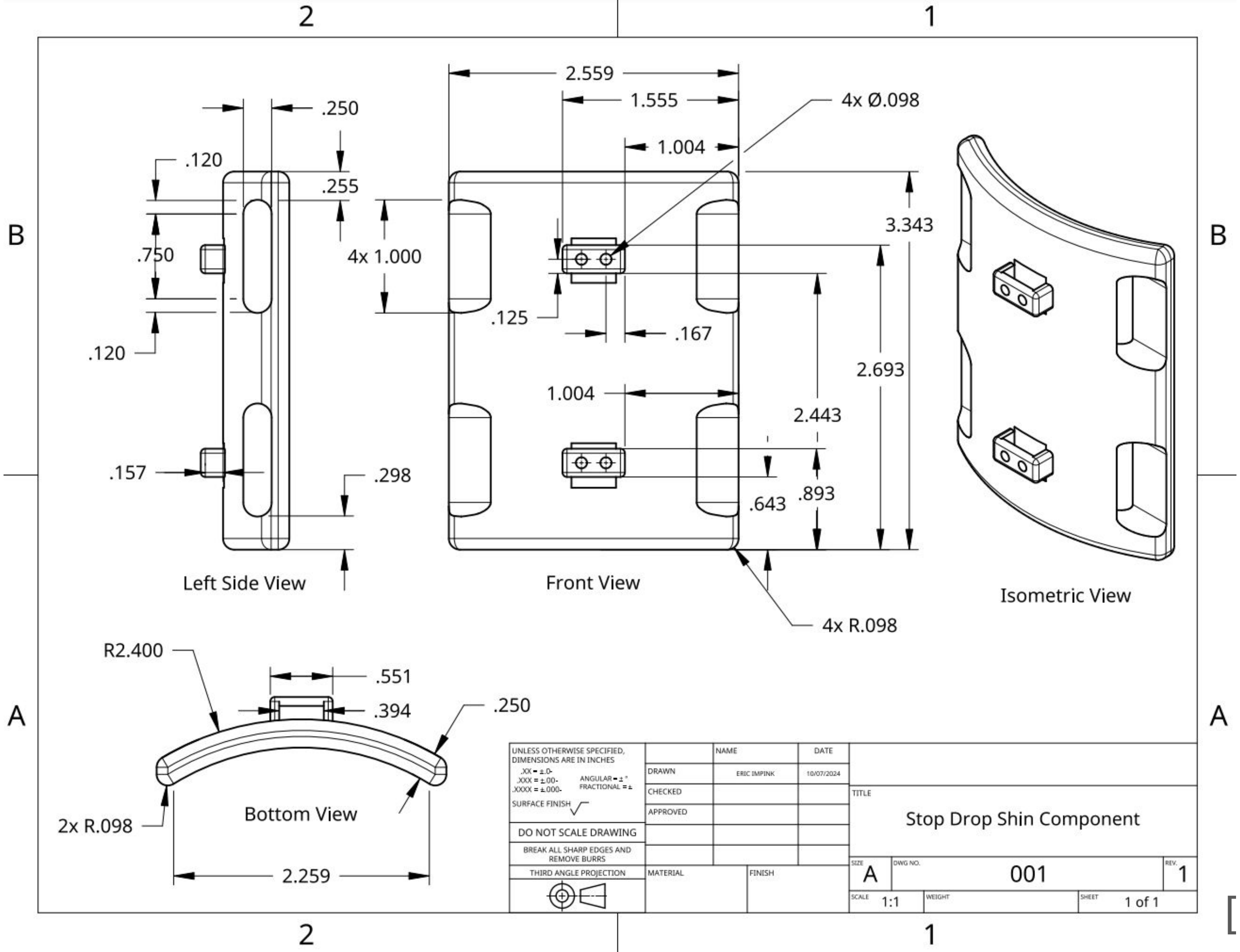


Padding

- Antimicrobial
- Lightweight

[12]

Not to relative scale



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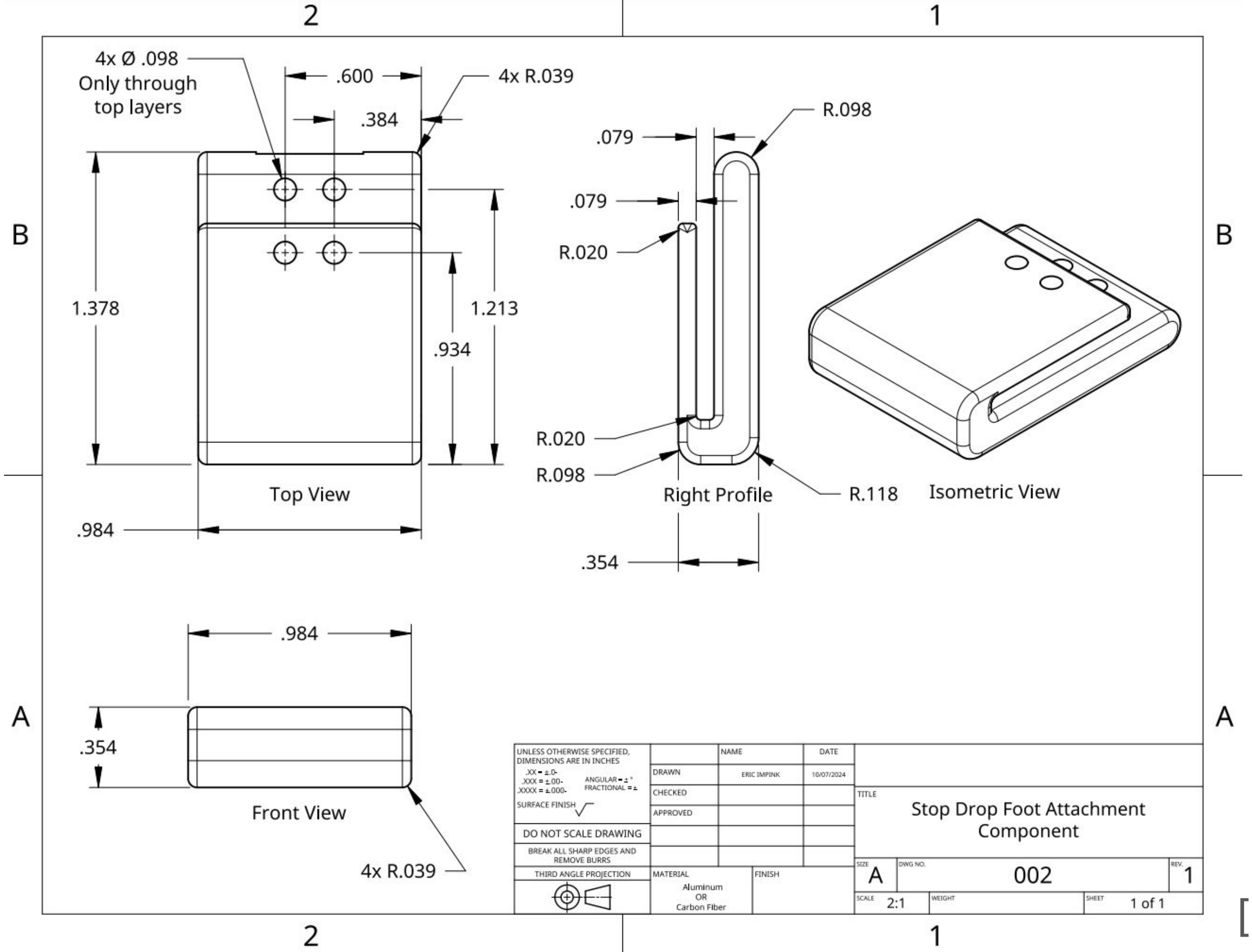
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[12]

Material Breakdown

Elastic / Bungee	Sourced from Amazon (joneaz) ~$3
Velcro	Sourced from personal materials
Padding	Sourced from Amazon(Talento) ~$10
Plastic (PLA)	- 3D printing setup and design, <$5 in materials + machine time. Near infinite customization. - PLA Filament= $18 - $25
Metal (future)	Most likely $100+ for metal 3D printing; (PCBWay, Protolabs, Xometry)

Financial Considerations

Current Device:

- Estimated Cost: ~$40-$60 USD

Future Device:

- Estimated Cost: $150+ (initially)
    - Price would decrease with:
        - Volume
        - Vendor relations
        - Material and Manufacturing Optimizations

Elastic / Bungee	- Bungee Cord - 250' / $85 = ~$3/foot ; Each device will probably use about a foot of cord. - Elastic (Exercise) Band - 75' / $50 = $1.50/foot ; Less durable, but cheaper.
Velcro	~$1 per strap x 3 (per device) = $3
Padding	~$10
Plastic	- 3D printing setup and design, <$5 in materials + machine time. Near infinite customization. - Injection molding - Large upfront cost, price per part would be minimal, limits customization.
Metal (future)	Most likely $100+ for metal 3D printing; - Redesign could make the device able to be manufactured in a sheet metal form or single punch for sub $100.

Regulatory Considerations (Device)

Medicare coverage

Medicare braces benefit
Social Security Act §1861(s)(9) [1]

- Be rigid or semi-rigid
- <u>Support a weak or deformed body part</u>
- Restrict or eliminate motion in a diseased or injured body part
- Be reasonable and necessary for the diagnosis or treatment of an illness or injury
- <u>Improve the functioning of a malformed body part</u>

Our device, in its current form, could be eligible for Medicare coverage, though further testing would be required.

FDA Approval

[Code of Federal Regulations]

[Title 21, Volume 8]

[CITE: 21CFR890.3475]

Sec. 890.3475 Limb orthosis. [2]

- "To support, to correct, or to prevent deformities or to align body structures for <u>functional improvement</u>"

FDA approval would require extensive testing, simulations, and documentation.

Regulatory Considerations (Business)

Quality Management Systems

ISO 13485:2016 [3]

- “Provide medical devices and related services that consistently meet customer and applicable regulatory requirements.”

Environmental management systems

ISO 14001:2015 [4]

- “A framework for organizations to design and implement an EMS, and continually improve their environmental performance.”

A strong organization will have a QMS. Will help build trust in our product.

EMS will build trust and support from environmentally conscious customer and investors.

Results and Conclusions

- The device was functional and successful per our initial design requirements.

- In testing, the device prevented tripping and enabled the 2MPH goal established at the beginning of our design phase.

- Statically, the device achieved a dorsiflexion angle of ~ 5° of dorsiflexion and ~30° of plantarflexion.

- Overall, the device provides a cost effective solution to active drop foot recovery.

Demonstration!

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